$$\frac{20y-6y}{15}=7$$

$$14y=7\times15$$

$$y=\frac{\cancel{7}\times15}{\cancel{14}_{2}}$$

$$y=\frac{15}{2}$$

Think



1. "An equation is multiplied or divided by a non zero number on either side." Will there be any change in the solution?
2. "An equation is multiplied or divided by two different numbers on either side". What will happen to the equation?

Exercise 3.6



1. **Fill in the blanks:**

   (i) The value of x in the equation $x + 5 = 12$ is ————.

   (ii) The value of y in the equation $y-9=(-5)+7$ is ————.

   (iii) The value of m in the equation $8m = 56$ is————.

   (iv) The value of p in the equation $\frac{2p}{3}=10$ is ————.

   (v) The linear equation in one variable has ———— solution.

2. **Say True or False.**

   (i) The shifting of a number from one side of an equation to other is called transposition.

   (ii) Linear equation in one variable has only one variable with power 2.

3. **Match the following :**

(a) $\frac{x}{2}=10$	(i) $x = 4$
(b) $20= 6x - 4$	(ii) $x = 1$
(c) $2x - 5 = 3 - x$	(iii) $x = 20$
(d) $7x - 4 - 8x = 20$	(iv) $x=\frac{8}{3}$
(e) $\frac{4}{11}-x=\frac{-7}{11}$	(v) $x = -24$

(A) (i), (ii), (iv), (iii), (v) (B) (iii), (iv), (i), (ii), (v)

(C) (iii), (i), (iv), (v), (ii) (D) (iii), (i), (v), (iv), (ii)

4. Find x and p (i) $\frac{2x}{3}-4=\frac{10}{3}$ (ii) $y+\frac{1}{6}-3y=\frac{2}{3}$ (iii) $\frac{1}{3}-\frac{x}{3}=\frac{7x}{12}+\frac{5}{4}$

5. Find x (i) $-3(4x + 9) = 21$ (ii) $20 - 2(5 - p) = 8$ (iii) $(7x - 5) - 4(2 + 5x) = 10(2 - x)$

6. Find x and m (i) $\frac{3x-2}{4}-\frac{(x-3)}{5}=-1$ (ii) $\frac{m+9}{3m+15}=\frac{5}{3}$

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